

Johansen Cointegration Test

Date: 07/09/26 Time: 11:59
Sample: 1/04/2010 12/30/2025
Included observations: 4159
Lags interval (in first differences): 1 to 4
Endogenous variables: GBPUSD CHFUSD
Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.002084	12.14983	15.49471	0.1499
At most 1	0.000839	3.485715	3.841465	0.0619

Trace test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.002084	8.664111	14.26460	0.3152
At most 1	0.000839	3.485715	3.841465	0.0619

Max-eigenvalue test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegrating Coefficients (norma

GBPUSD	CHFUSD
1.330499	14.26505
6.621978	-3.801507

Unrestricted Adjustment Coefficients (alpha):

D(GBPUSD)	-4.73E-05	-0.000218
D(CHFUSD)	-0.000298	-5.46E-05

1 Cointegrating Equation Log-Likelihood: 29587.28

Normalized cointegrating coeffici

GBPUSD	CHFUSD
1.000000	10.72158
	(3.76489)

Adjustment coefficients (standard

D(GBPUSD)	-6.30E-05	(0.00016)
D(CHFUSD)	-0.000397	(0.00014)